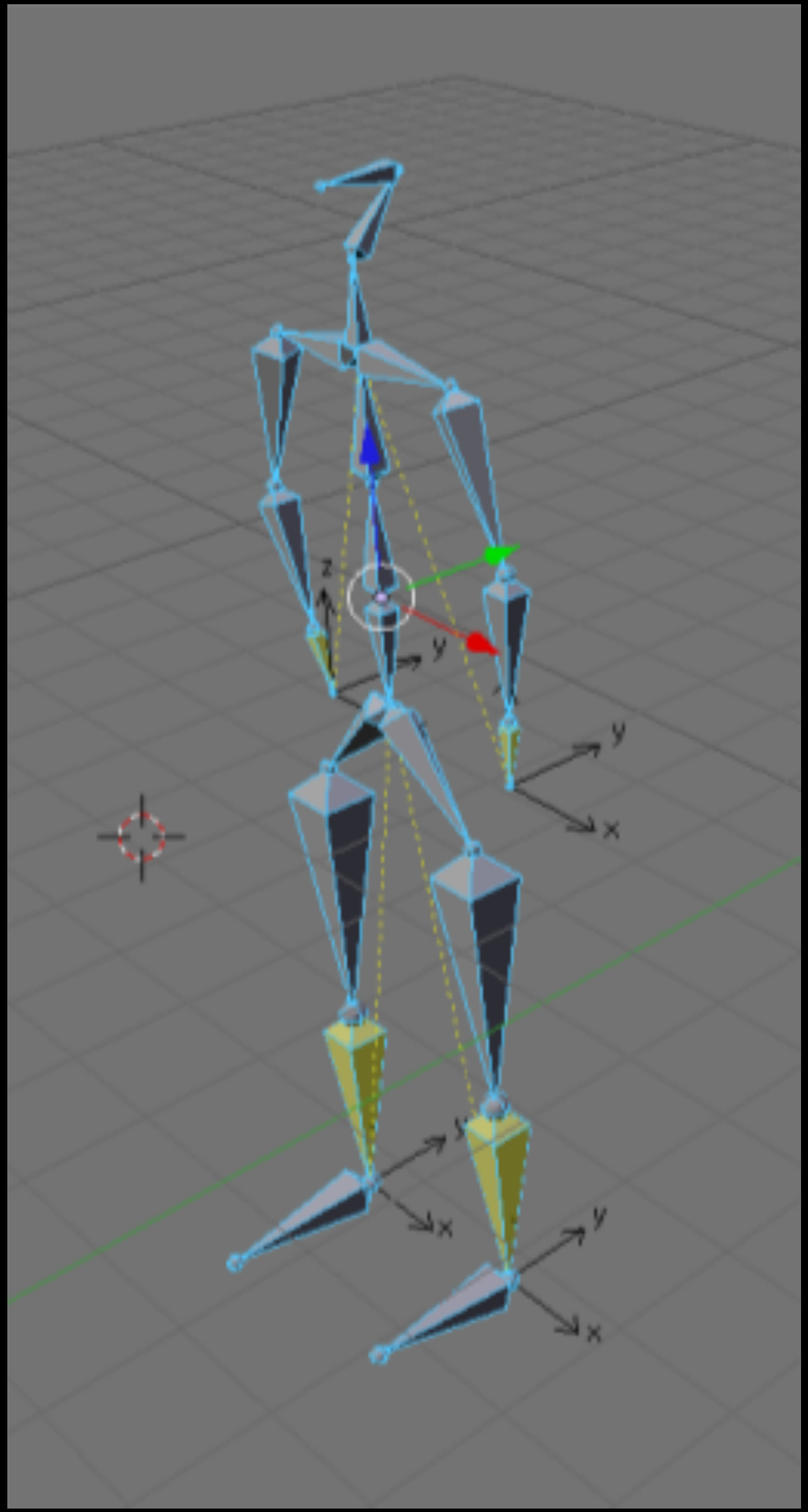
Advanced Computer Graphics

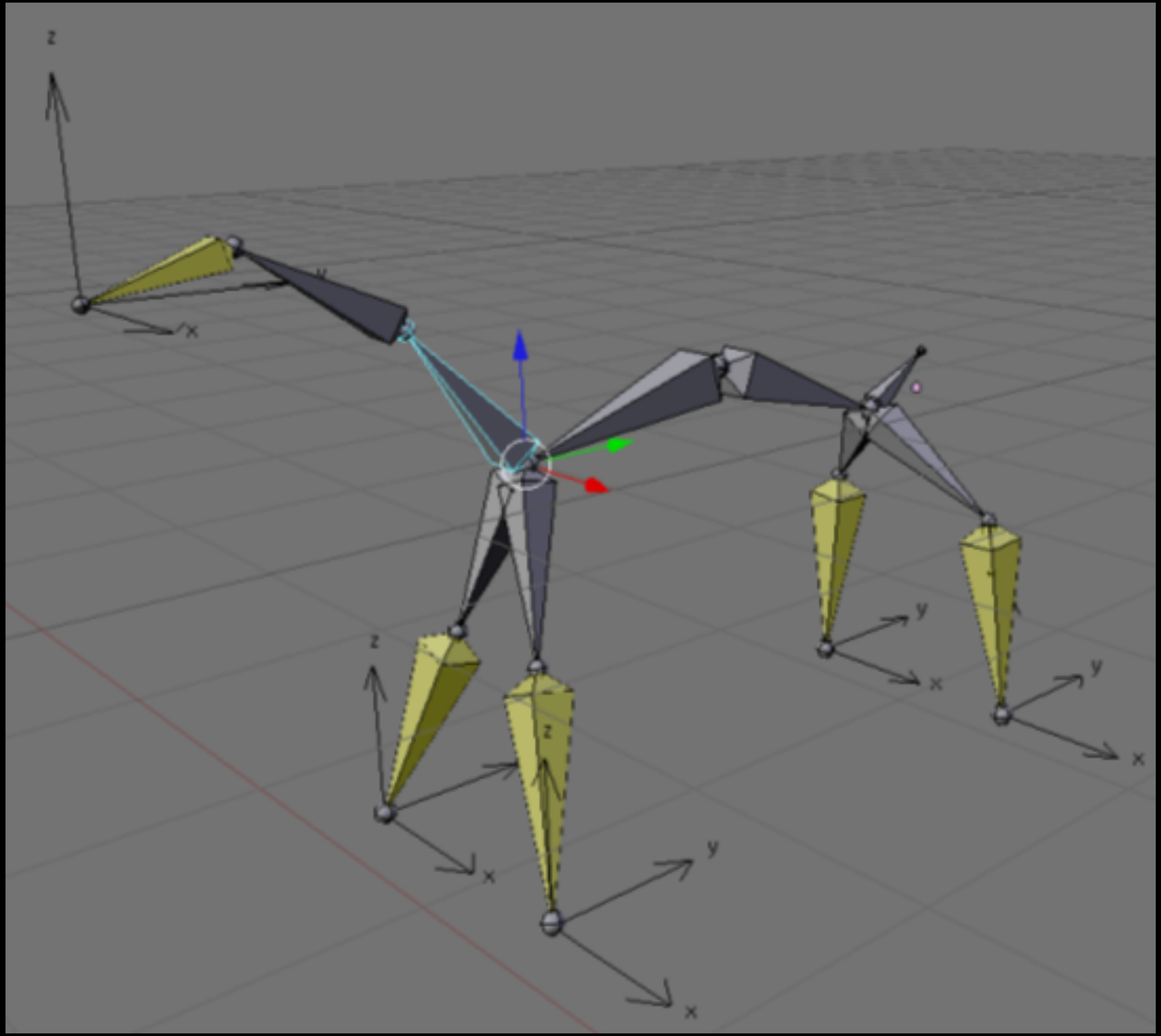
Motion Retargetting

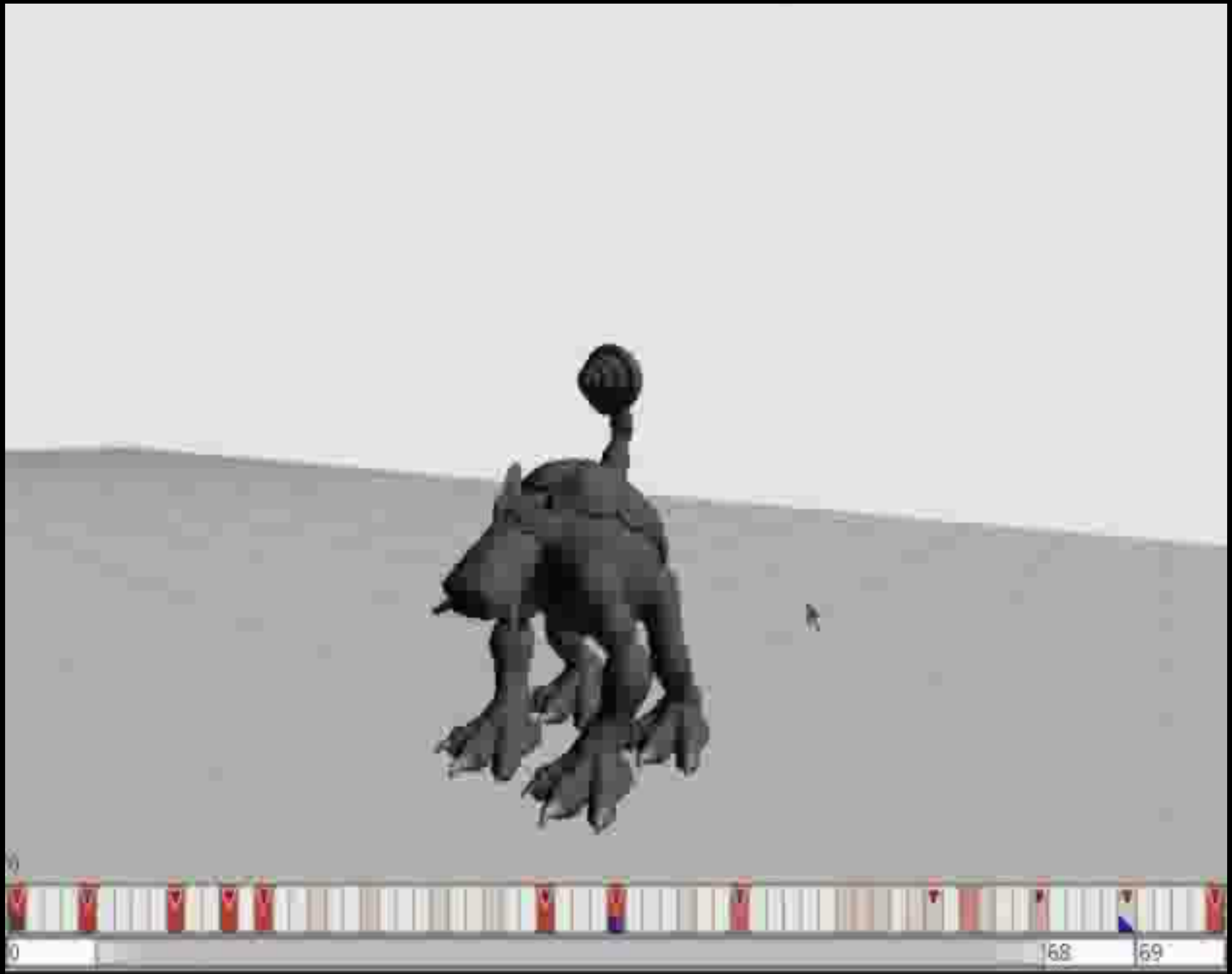
What is Motion Retargeting?

- A method to retarget animations onto models with different morphologies.

- A way to remap animations onto characters with very different animation-specific structures.



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Motivation for Motion Retargeting

- Improves content reuse
- Allows for a more dynamic game experience
- Easy integration of procedurally generated animations

- Possibility to create exciting new game concepts (example: Spore)

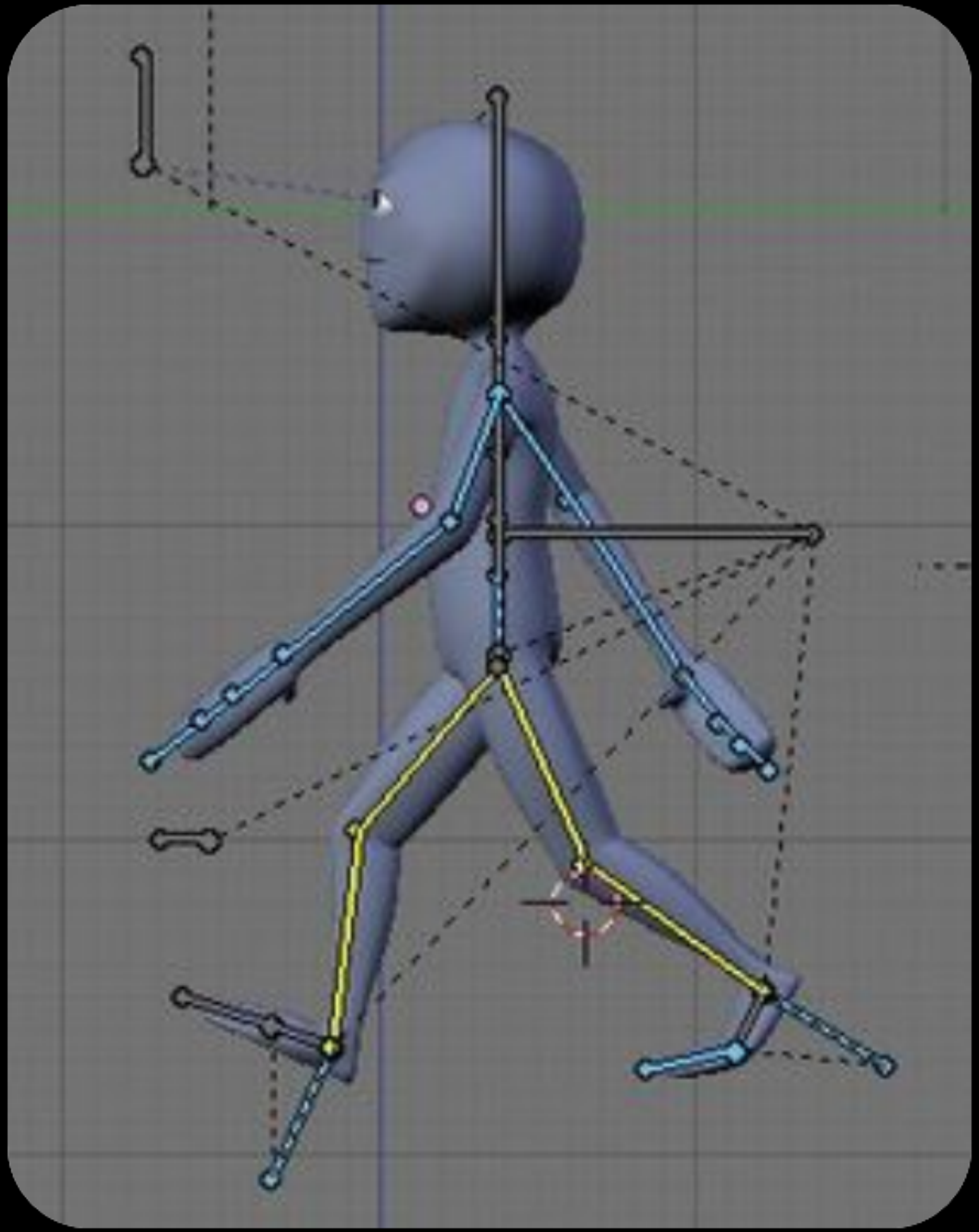
Conventional Animation Authoring

Animation Authoring consists of 4 steps:

- Selecting the bones
- Moving the bones into a new pose
- Saving a key frame
- Playback to locate issues

Rig and Model!

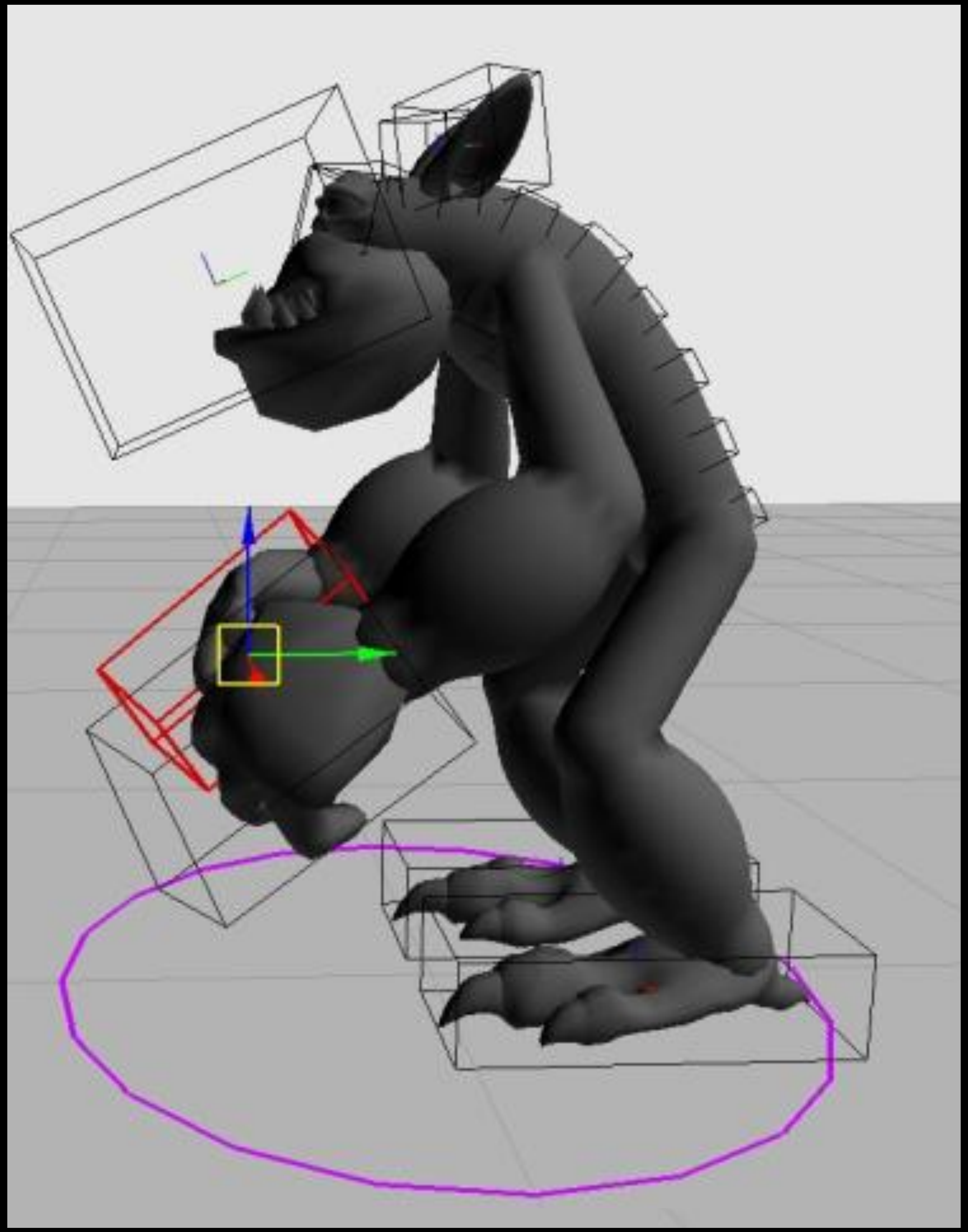
What if we don’t know about them?



Animation Models

Models are made up of:

- bodies
- meshes
- textures



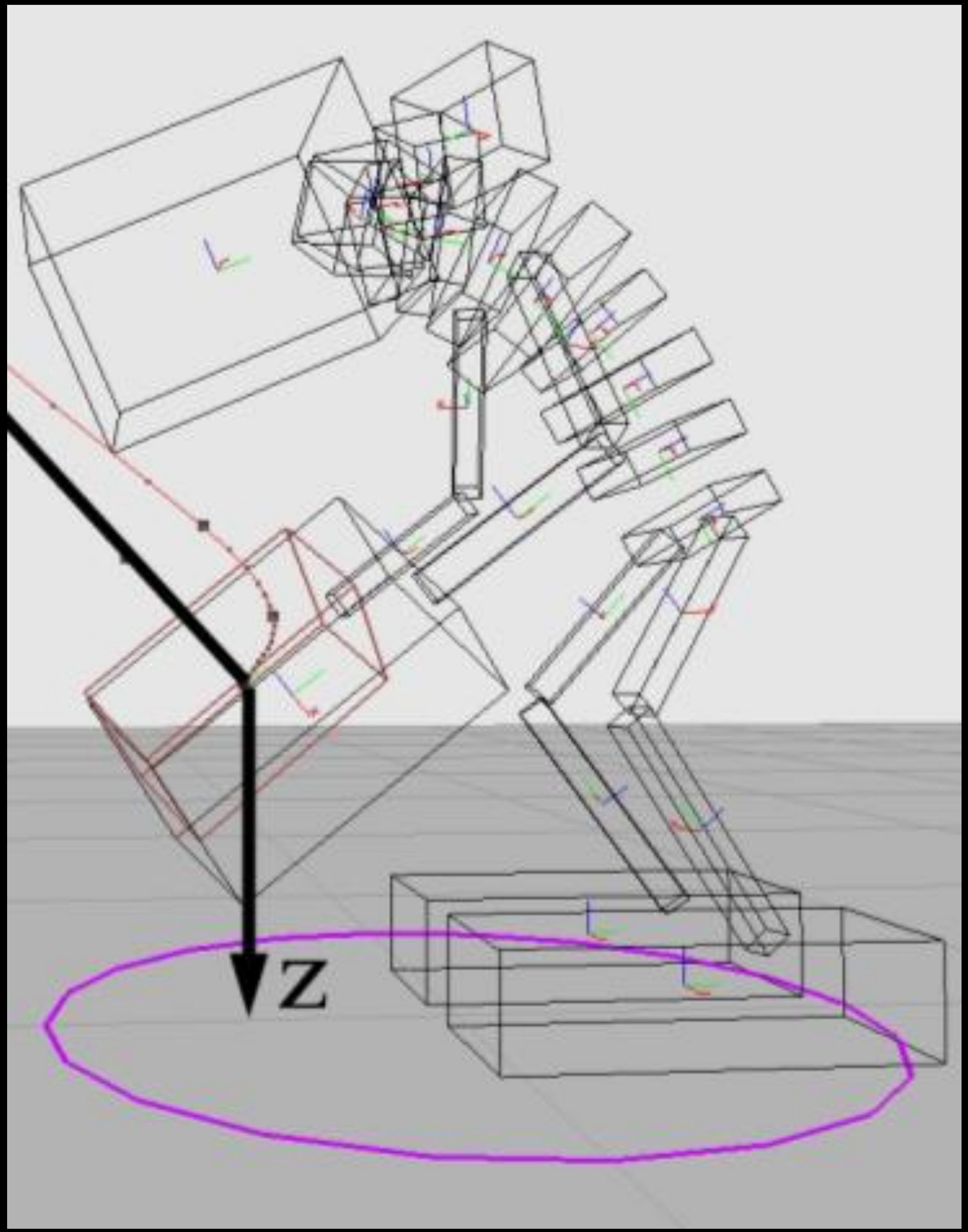
Animation Models Examples

- Bodies consist of:
  - position
  - orientation
  - functions
  - etc.

- Meshes are attached to bodies.

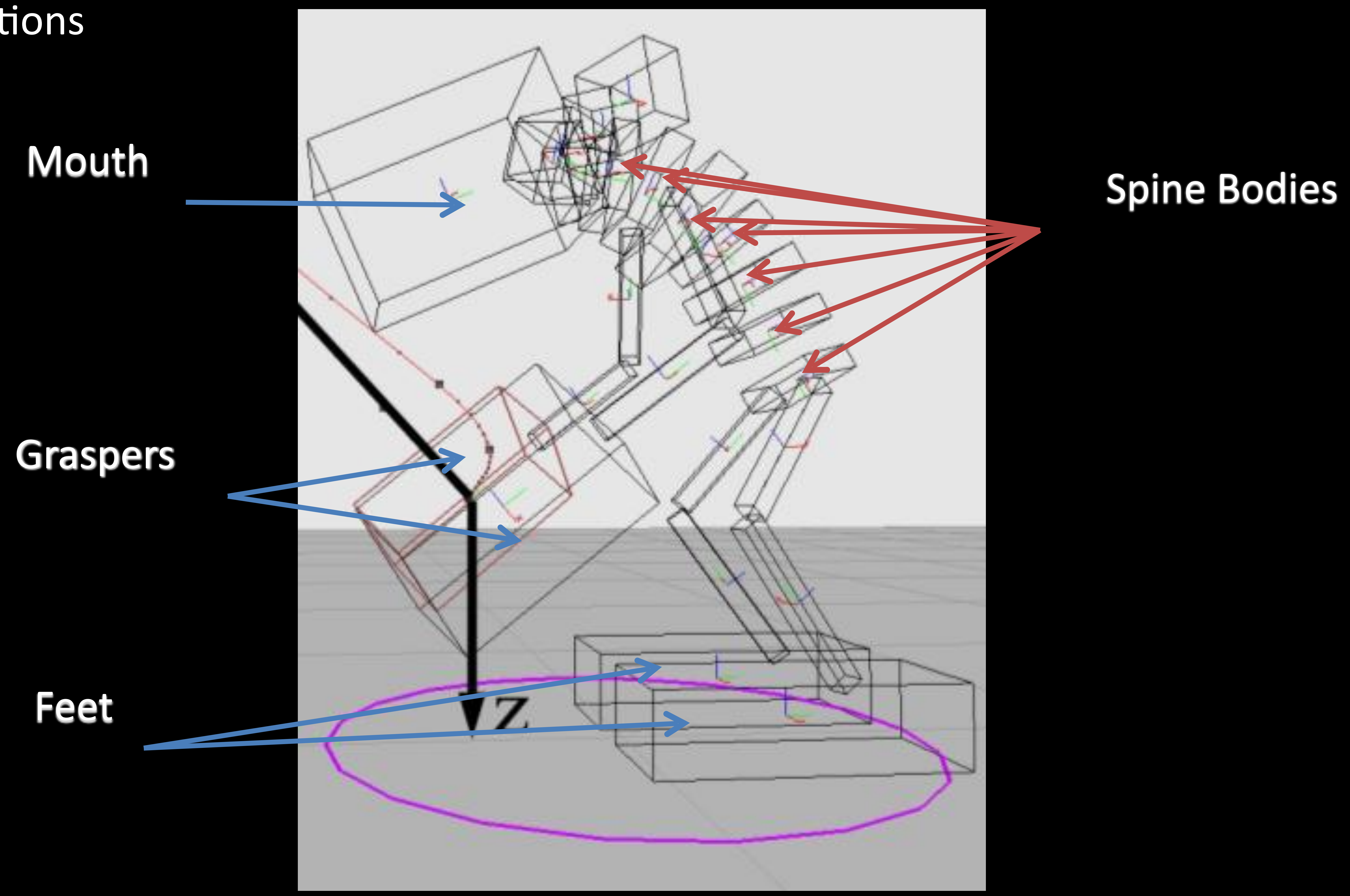
Animation Models

- The bodies form a DAG with linkages from the root
- Root is at the pelvis, usually
- The pose, a body is created in is known as **rest pose** or **standard pose**

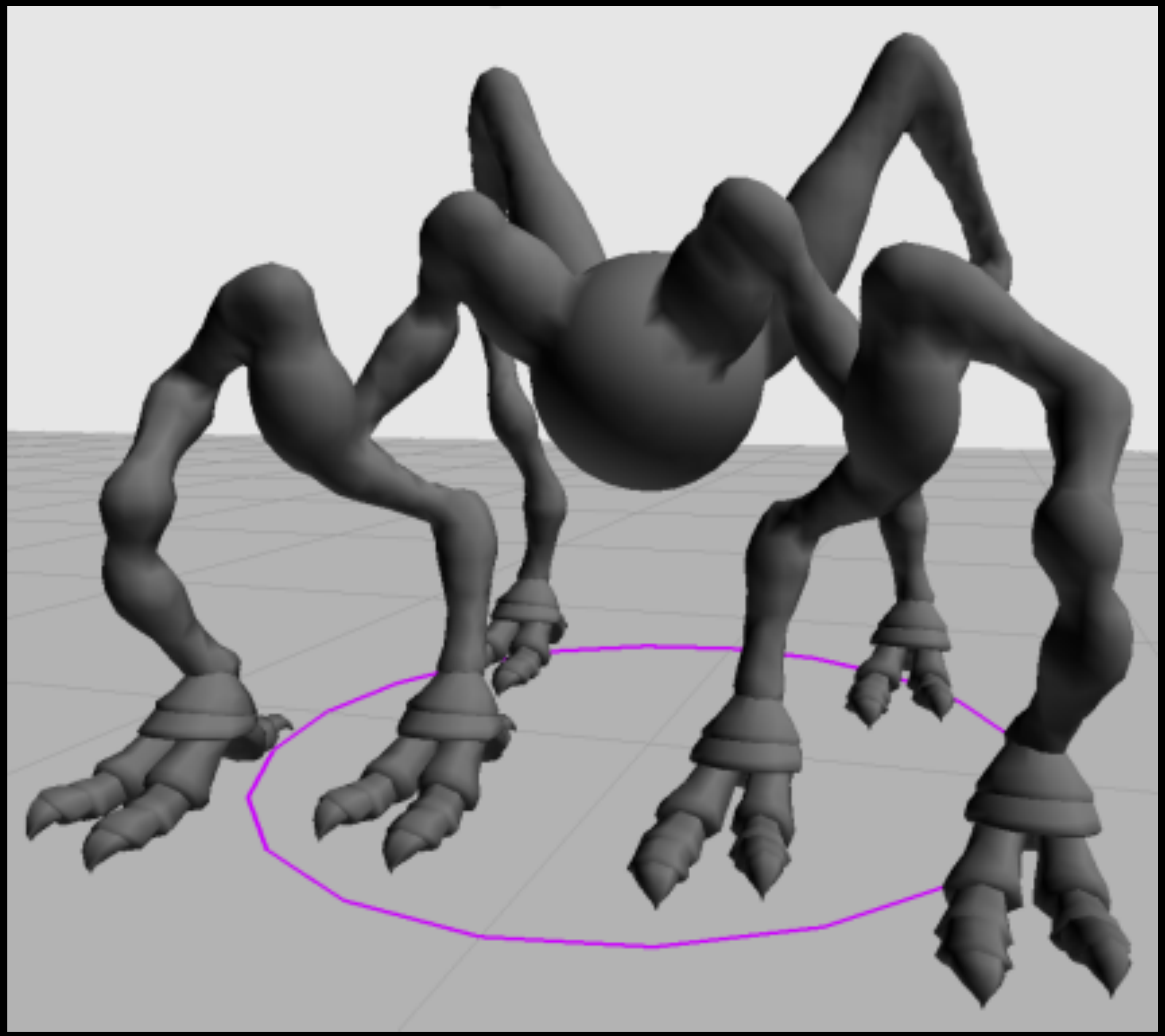


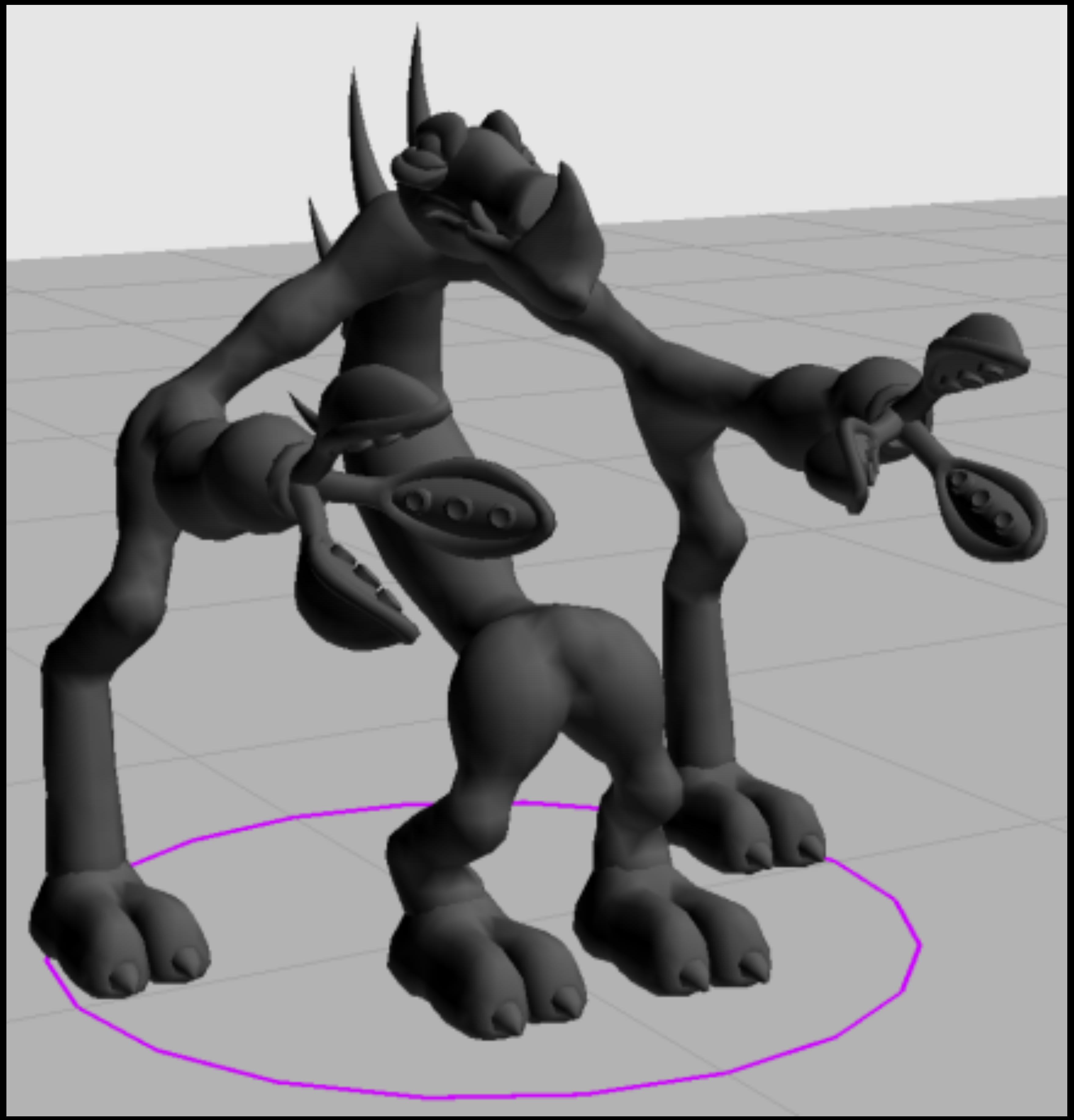
Example

Functions



Two More Complex Examples





Step 1: Selection

- Select links from the original body to target body

- How to find?
  - Mechanical findings
  - Find by semantic methods

Step 1: Selection

Selection Query would contain

- functions
- spatial query
- extent query
- Etc.

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What is the type of the body:

- mouth
- graspers
- spine
- root

Step 1: Selection

Selection Query would contain

- functions
- spatial query
- extent query
- Etc.

Where are the bodies:

- Front/Center/Back
- Left/Center/Right
- Top/Center/Bottom

relative to:

- the character
- other bodies with the same caps

Step 1: Selection

Selection Query would contain

- functions
- spatial query
- extent query
- Etc.

Select the body that is:

- FrontMost/BackMost
- LeftMost/RightMost
- TopMost/BottomMost

Step 1: Selection

Selection Query would contain

- functions
- spatial query
- extent query
- Etc.

Step 2: Posing

Specify a pose for the bodies for selected links

Step 2: Posing

Generalize the posing process by:

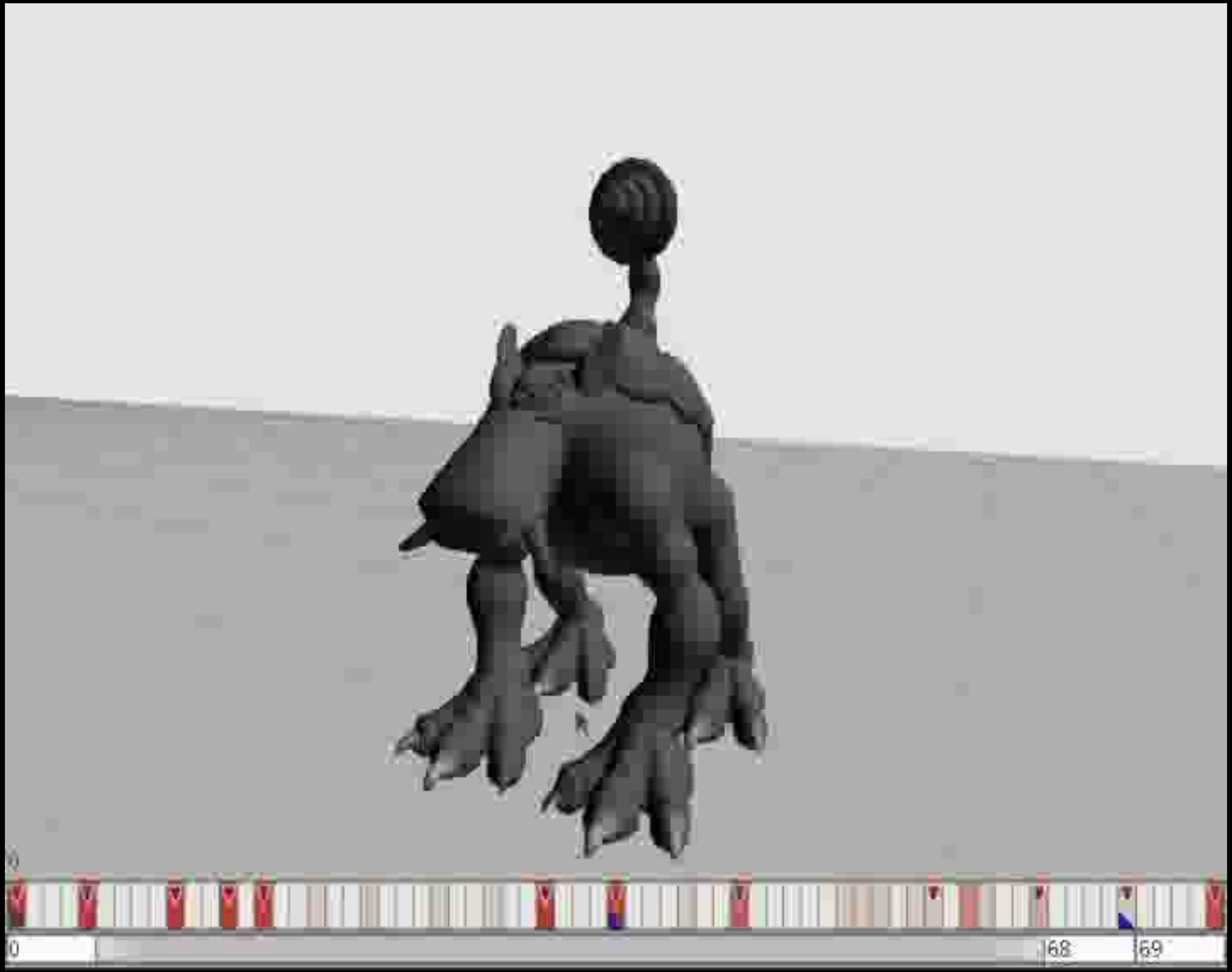
- Providing different movement modes
- Applying changes from the active body to all bodies according to the movement mode

Note:

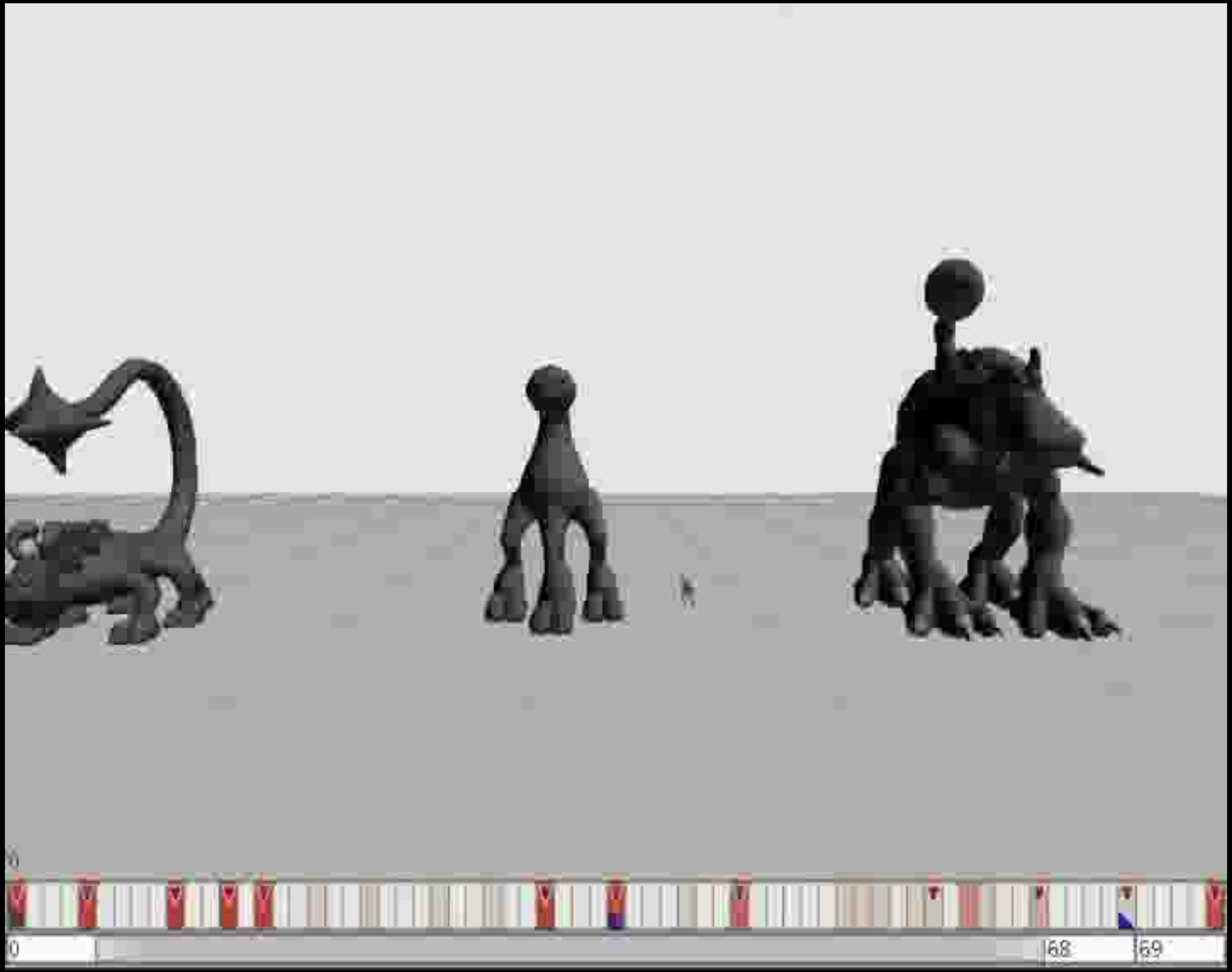
Channels animate specific DoFs only:

→ Rotation and position can be animated by different channels

Posing Examples



Posing Examples



Realization

Active body is moved

Generalization of the movement

Generalized movement is stored

Specialization onto all selected bodies

Selected bodies are moved accordingly

Movement Modes

Movement Modes:

- Identity
- Rest Relative
- Scale Mode
- Ground Relative
- Secondary Relative Movement
- Lookat
- Mirroring Mode

Movement Modes

Movement Modes:

- **Identity**
- Rest Relative
- Scale Mode
- Ground Relative
- Secondary Relative Movement
- Lookat
- Mirroring Mode

Absolute movement mode. Bodies are assigned the same position and orientation.

Movement Modes

Movement Modes:

- Identity
- **Rest Relative**
- Scale Mode
- Ground Relative
- Secondary Relative Movement
- Lookat
- Mirroring Mode

Movement is relative to the orientation and position of the body in its respective rest pose.

Movement Modes

Movement Modes:

- Identity
- Rest Relative
- **Scale Mode**
- Ground Relative
- Secondary Relative Movement
- Lookat
- Mirroring Mode

Scaling of the relative movement depending on:

- CreatureSize
- LimbLength

Movement Modes

Movement Modes:

- Identity
- Rest Relative
- Scale Mode
- **Ground Relative**
- Secondary Relative Movement
- Lookat
- Mirroring Mode

Changes one axis to be perpendicular to the ground.

Scales it so 0.0 is the height of the body in the rest pose and 1.0 is on ground level.

Movement Modes

Movement Modes:

- Identity
- Rest Relative
- Scale Mode
- Ground Relative
- **Secondary Relative Movement**
- Lookat
- Mirroring Mode

An ExternalTarget is chosen and an axis is aligned with the distance vector of the external target and the body in a way that 0.0 is at the rest position of the body and 1.0 is at the target.

We can also only use the direction in SecondaryDirectionalOnly mode.

Movement Modes

Movement Modes:

- Identity
- Rest Relative
- Scale Mode
- Ground Relative
- Secondary Relative Movement
- **Lookat**
- Mirroring Mode

Rotates the body towards the ExternalTarget and applies soft constraints on the rotation if it the object "is behind" the body.

Movement Modes

Movement Modes:

- Identity
- Rest Relative
- Scale Mode
- Ground Relative
- Secondary Relative Movement
- Lookat
- **Mirroring Mode**

Mirrors the movement on an automatically generated sagittal mirror plane depending on which side of the plane the body is.

Step 3: Blending & Keying

- Animators can define **blend groups** for each channel to control the way how different animations are combined or how the transitions are done
- The generalized pose data could be saved into curve parameters

Step 4: Animation Instantiation

- Generalized animations are retargeted onto models.

- Two features to control the variety of possible animations:
  - branching
  - variants

Branching

- It’s not always possible to create animations that fit all imaginable characters

→Add branch predicates to animations to filter the possible animation set

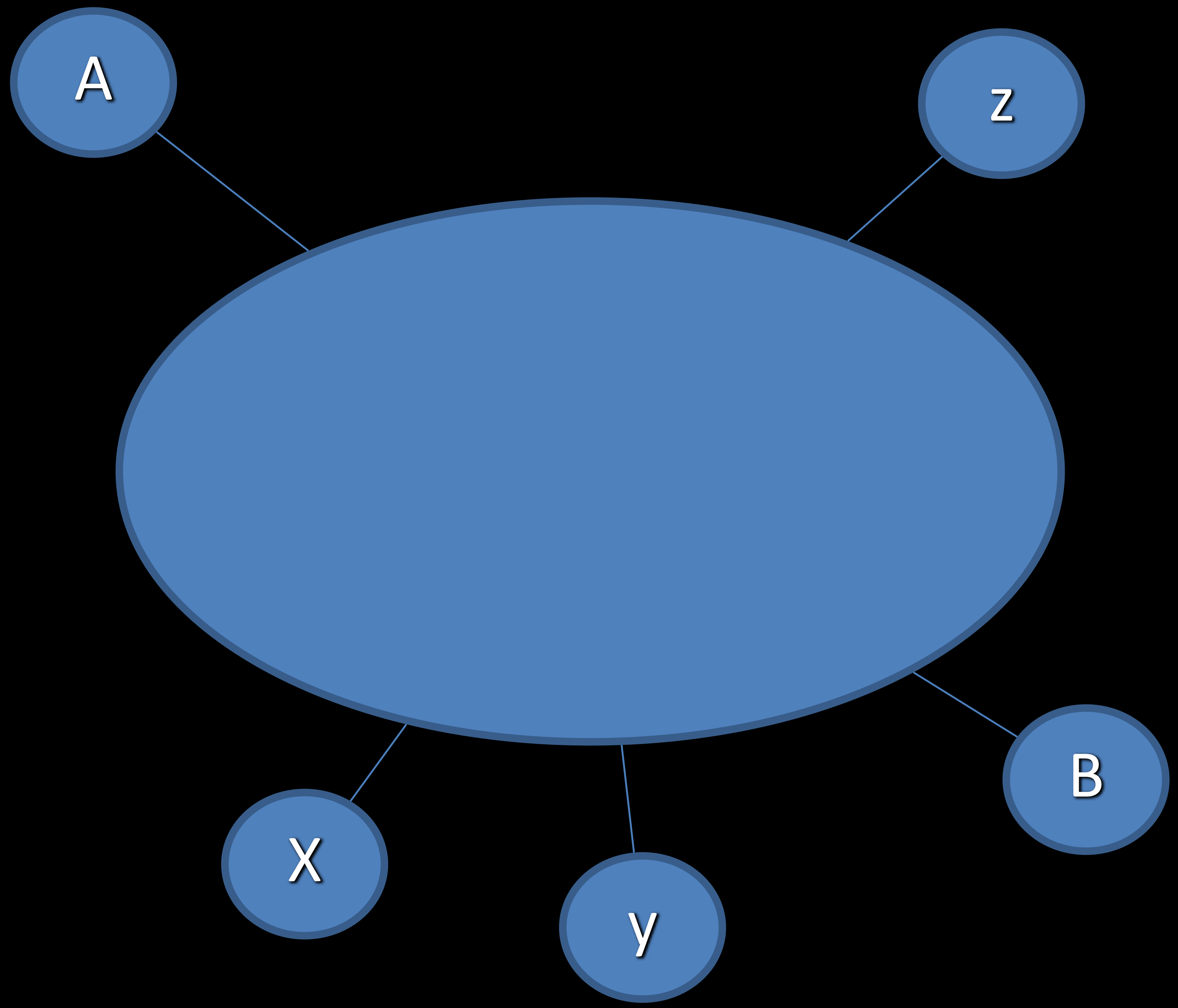
For example:

- UprightSpine
- HasGraspers
- HasFeet

Variants

- How many ways are there to play an animation on a character?
- “**Select Only One**”-option for channels can add lots of combinations
- Possibility to group channels in variant groups
- Put constraints on the spatial relationship between channels in variant groups

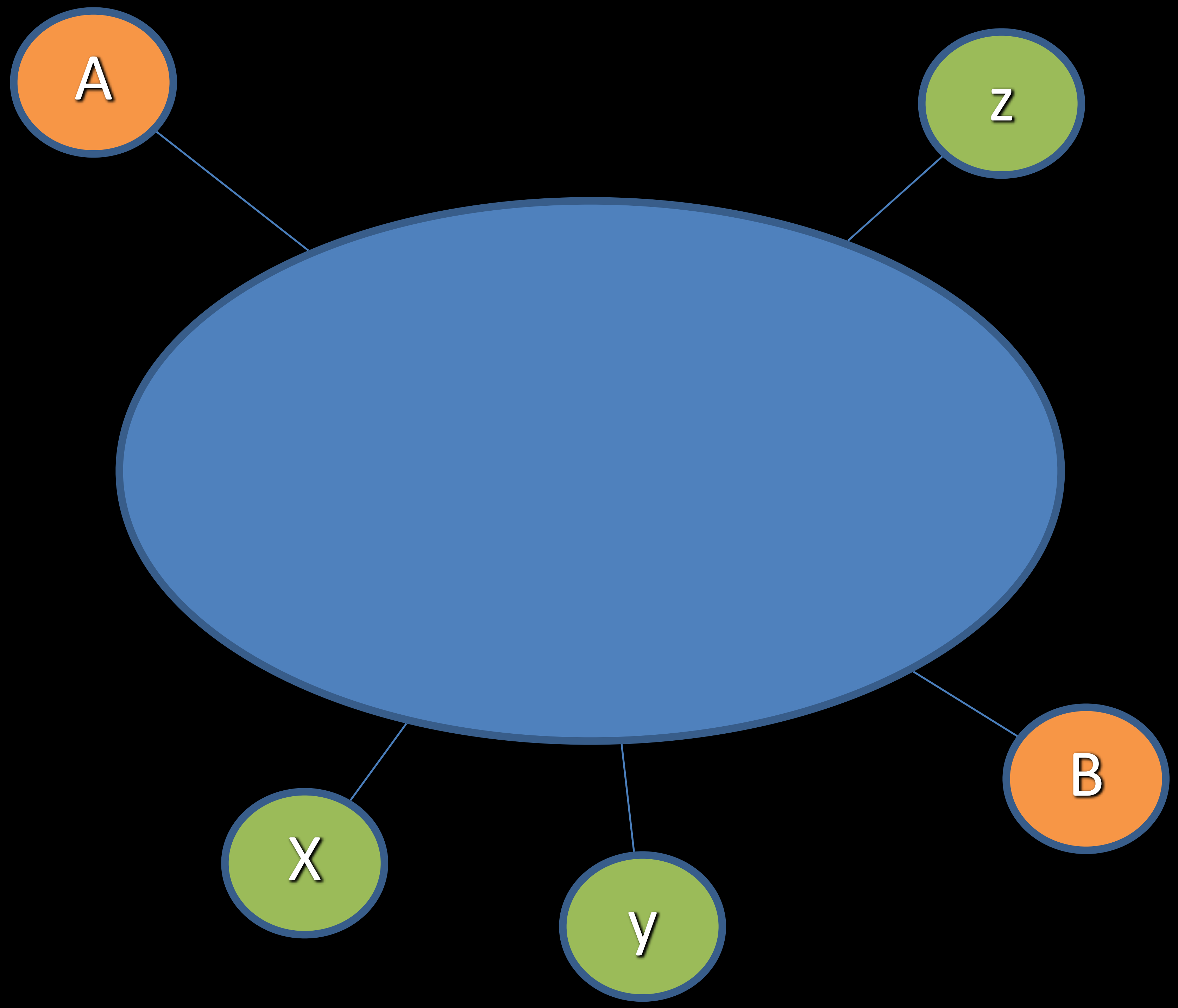
Variants (Example)



Possible Combinations (animation that accesses one body):

A B x y z

Variants (Example)

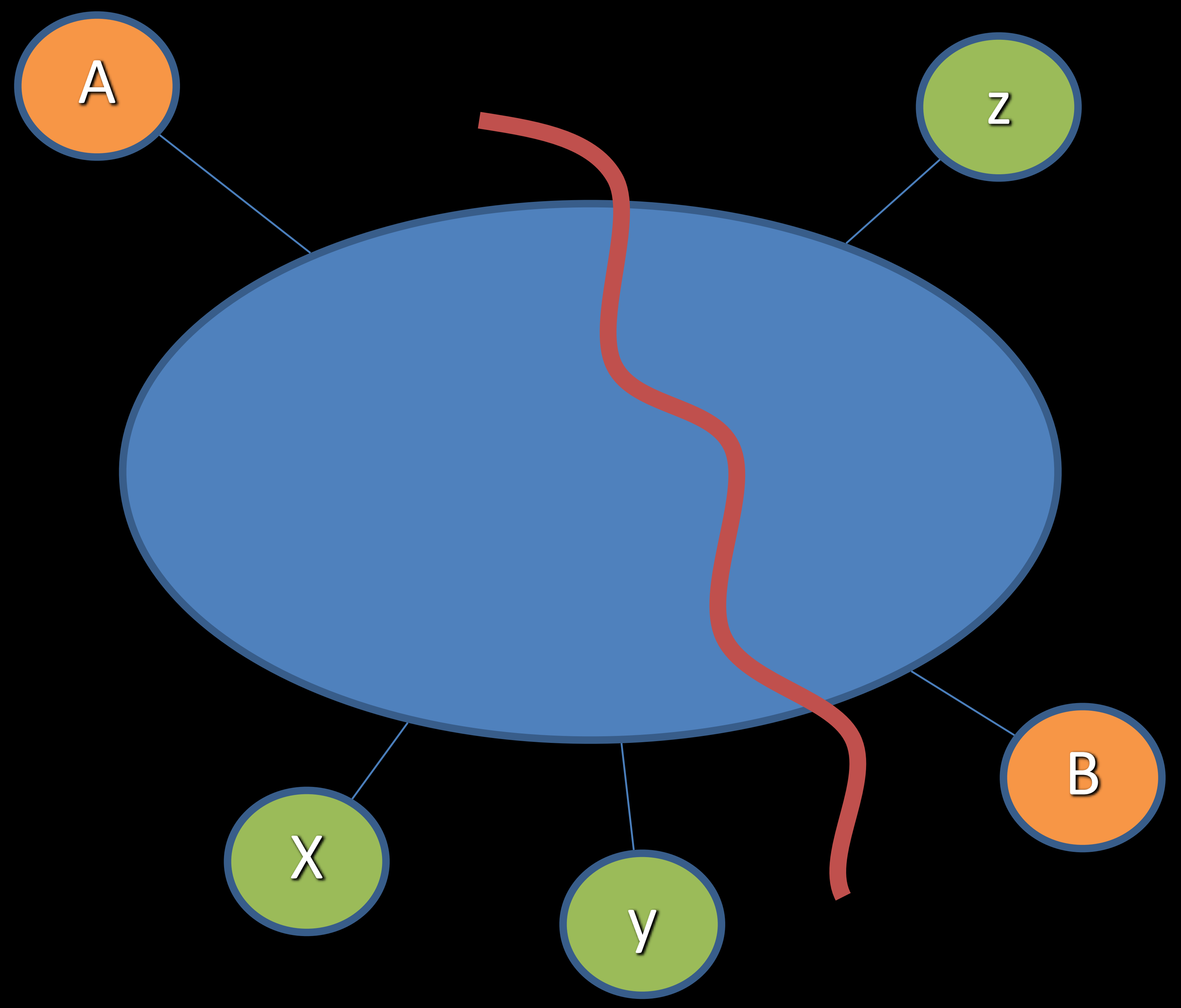


Possible Combinations (animation that accesses one of each):

Ax Ay Az

Bx By Bz

Variants (Example)

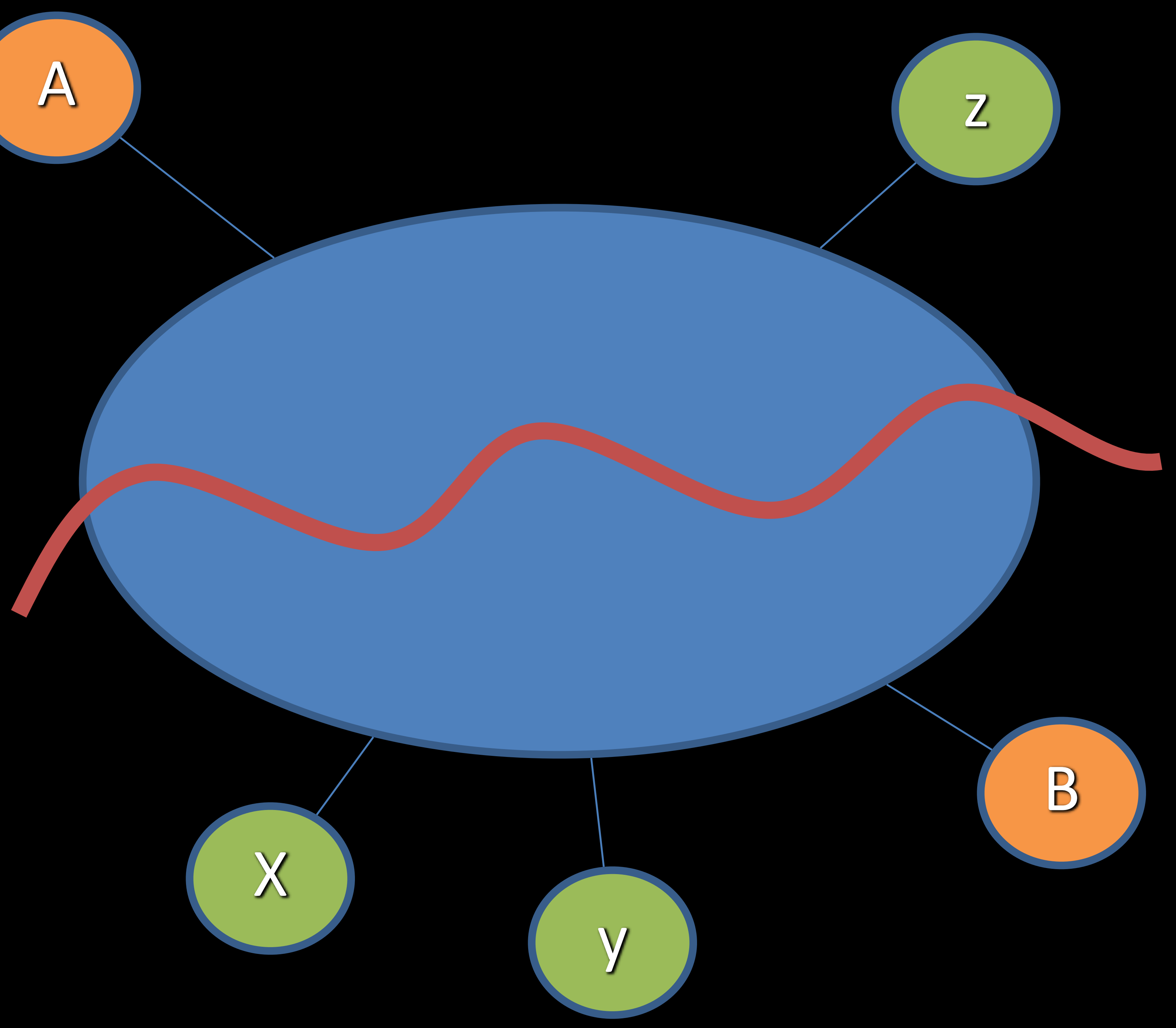


Possible Combinations (opposite-side pairs only):

Ax Ay

Bz

Variants (Example)



Possible Combinations (same-side pairs only):

Az

Bx By

Gaits

A gait is a pattern of locomotion characteristic of a limited range of speeds, described by quantities of which one or more change discontinuously at transitions to other gaits.

- Characters also need to walk and run, that is be able to move in a convincing way

- If characters have no legs, they still need a way to move:
  - by crawling
  - by floating

Gaits

- Gaits can often be generated procedurally and still look good
- In fact, you may need to generate part of them dynamically because one can’t guess the leg configurations, etc.

Gaits

- Gaits are cyclic
- Leg movements are cyclic, too
- The **duty factor** for a leg describes the period of time (percentage of the total gait cycle length) in which the foot is on the ground
- The **step trigger** is the time (normalized) in a gait cycle on which the foot leaves the ground
- Other definitions possible but similar usually

Gaits (Example)

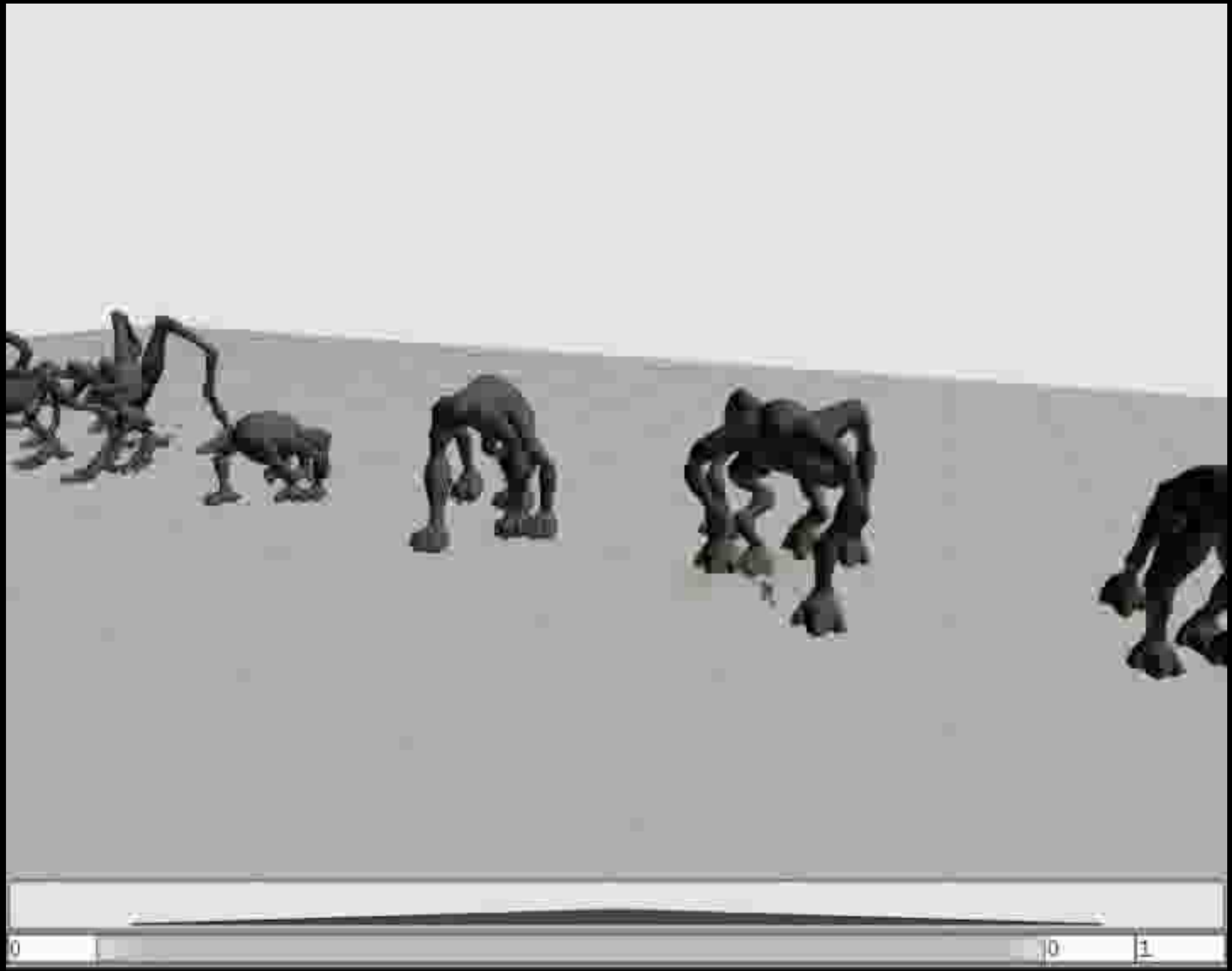
Biped Walk

Duty Factor: .6 ~ .8

Quadruplet Walk

Duty Factor: .6 ~ .8

Gaits Example



Conclusion

- Emphasize content reusability and the benefits of adding more semantic details to animations and assets in general during authoring.
- “It’s easy to add semantic attributes for you, but it’s incredibly hard (for the computer) to figure it out later.”

Conclusion

- Always try to simplify the problem you work on
- Complicated solutions are often too rigid to be practical
- Iterate your design and code a few times to find the best implementation